AI FOR SCIENCE RESEARCHER · FOUNDER

Jiaxuan Chen 陈嘉轩

I work across physics, scientific computing and the evaluation of AI systems. My interests include particle phenomenology, gravitational-wave inference, mathematical physics and statistical identification.

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01 ABOUT AND RESEARCH PRACTICE

I hold an Sc.M. in Physics from Brown University and a B.S. in Space Science and Technology from Shandong University. Research experience spans high-energy physics, thermal transport, nuclear spectroscopy and astrophysical data analysis.

Alongside research, I founded Omytea to develop personal-agent systems, scientific work environments and tools for understanding uncertain futures. I also work on motion-analysis products and write about technology, economic power and life after abundance.

AI-assisted research. AI tools support mathematical exploration, code development, computational analysis, checking and manuscript preparation. Results and validation scope are stated for each project. AI for Science describes the workflow; it does not mean every project develops an AI model. Original research periods and later AI-assisted work are recorded separately.

02 EDUCATION

Brown University — Sc.M. in Physics

2024 — 2026

Providence, Rhode Island, USA. Degree awarded 23 May 2026. Graduate research in CMS jet simulations and carbon-nanotube 3ω thermal characterization.

Shandong University — B.S. in Space Science and Technology

2020 — 2024

Degree awarded 21 June 2024. Undergraduate work in X-ray binary and stellar-flare astrophysics.

Stages are labelled individually. A submitted manuscript is distinct from a public preprint, an accepted paper or a published article.

Coherent Strange-Current Contributions to X(2370) → ηηη

2026.08 · Particle phenomenology Working manuscript · October 2026

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Analyses coherent strange-current contributions to X(2370) → ηηη, tracking amplitudes, interference and phase-space factors rather than replacing the coherent sum with separate rates.

At the stated reference inputs, the light and strange amplitudes interfere destructively; the completed ηηη width is smaller by a factor of approximately 30.88. The charge-complete reference ratio Γ(ηηη)/Γ(KKπ) is approximately 3.315×10^{-4}. The sensitivity box contains a cancellation region; this is a conditional channel prediction, not a measurement of glueball composition.

Related earlier work examined two-photon production and Belle constraints. Amplitude ambiguity prevented the proposed broad-family limit; that study is retained as an archived negative result, distinct from the current strange-current manuscript.

Methods, related work and links ↗

Convergence of local modular data for half-line free fermions and a locality/passband obstruction to raw fixed-cutoff global quasi-equivalence

2024.05–2024.08 · Mathematical physics Public preprint · Version 2, 31 July 2026

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Studies the continuum limit of finite-dimensional modular data for half-line free fermions using compact-local wavelet schemes. Under the stated hypotheses, local restrictions converge in norm and the modular generators converge in the strong-resolvent sense after faithful block completion.

A locality/passband obstruction distinguishes local convergence from raw fixed-cutoff global quasi-equivalence. Version 2 corrects the earlier endpoint-rate discussion; current external journal-review status is not asserted.

Methods, related work and links ↗

Formation Mechanism of Blazar Jets

2022.07–2023.01 · Astrophysics · UCL research project Published paper · 2023

A single-author study of blazar-jet formation, comparing jet power, accretion-disc luminosity and the physical mechanisms associated with Blandford–Payne and Blandford–Znajek models. Published in Theoretical and Natural Science, volume 19, pages 169–175 (2023).

Methods, related work and links ↗

What Public AI Compute Records Can Establish: Evidence, Access and Comparison

2026.01–2026.07 · Technology policy and documentary research

Submitted manuscript · IJTPM, October 2026

A bounded documentary study of six public AI-compute programmes, asking what published records can establish about access, allocation and comparison.
The analysis separates programme-level announcements, accounting units, tariffs and evidence for named-project consumption. It does not infer confidential usage, causal policy failure or the absence of activity beyond the documentary frame. Last verified journal stage: screening.

Methods, related work and links ↗

Strongly supercooled transition dynamics and tachyonic modes

2025.09; comparison 2026.10 · Particle theory and transition dynamics

Unpublished computational and theoretical research

AI for Science · AI-assisted

Studies strongly supercooled transition dynamics using an RG-improved potential, an O(4) bounce, Euclidean initialization of fluctuations and subsequent mode evolution in Milne coordinates. Original research dates to September 2025; the documented comparison was performed on 16 October 2026.

Four conditional trajectories compare the numerical evolution under specific initialization and backreaction assumptions. Shooting/collocation agreement and equation residuals check bounded numerical steps; physical-consistency questions concerning the vacuum and Ward constraints remain open.

Related studies address spillway-preheating inverse problems, typed tachyon representations, order parameters and finite-window phase-boundary criteria. These are narrower working lines, not a universal completed classification or an independently established new phase transition.

Methods, related work and links ↗

Calibration diagnostics for ringdown overtone tests

2024.06; analysis 2026 · Gravitational-wave inference

Working manuscript · analysis in progress

AI for Science · AI-assisted

Studies how start time, conditioning, processing choices and null calibration affect the significance assigned to a black-hole ringdown overtone. The programme began in June 2024; the computations described here were carried out in 2026.

A controlled synthetic Gaussian-mixture null study used 199 calibration and 200 test samples. The conditioned test rejected 10/200 null samples (5%; 95% binomial interval 2.42–9.00%). This bounded synthetic diagnostic does not establish coverage in real detector noise.

Related work compares waveform and posterior conventions, including a separate GW250114 posterior audit. The illustration below is an explanatory visualization, not a new detector observation or event-discovery result.

Methods, related work and links ↗

Population-feature significance and finite selection simulations

2026 · Population inference and statistical diagnostics Unpublished analysis and manuscript

AI for Science · AI-assisted

Audits finite-injection uncertainty for population-feature claims in GWTC-4, relating catalogue events, selection simulations and feature-local effective sample size.

The proposed explanation based on severe local underconvergence was not supported by the diagnostic. The work also studies bounded finite-selection null simulations; it does not establish a universal distribution-free significance certificate.

Methods, related work and links ↗

GW250114: posterior conventions and exclusion diagnostics

2026.06 · Gravitational-wave inference Unpublished posterior reanalysis

AI for Science · AI-assisted

Compares seven public GW250114 posterior sets, examining start-time choices, amplitude conventions and how these enter simplified overtone-exclusion calculations.

Simplified exclusions span 0.91σ–3.45σ under the examined choices. This is a posterior-convention audit within the ringdown programme, not a complete new parameter-estimation analysis or a discovery claim.

Methods, related work and links ↗

Digit Dependence under Translation-Conjugated Cyclic Rotation

2024.05–2024.08 · Discrete mathematics and computation Unpublished manuscript

AI for Science · AI-assisted

Studies how translation-conjugated cyclic rotation changes the dependence of output digits on input digits. The manuscript constructs a complete pivotality matrix under an explicit finite-register model.

The analysis relates dependence to carry structure and establishes a bounded-fan-in circuit-depth result within the stated model. Finite enumerations check examples; they are not substituted for the general argument.

Methods, related work and links ↗

Finite-register locality and machine-dependent ranking

2024.05–2024.08; manuscript 2026.09 · Mathematical foundations and computation

Unpublished working manuscript

AI for Science · AI-assisted

Studies explicit finite-register representations, locality, short descriptions and the dependence of representation rankings on the chosen machine.

The claims concern conditional description-length comparisons and specified ranking constructions. They do not establish an impossibility theorem for every universal prior or a universal disproof of Born-rule derivations. Related work derives the digit-dependence structure of translation-conjugated cyclic rotation.

Methods, related work and links ↗

▶ Observer-state foundations: manuscripts and research history

05 AI, SCIENTIFIC SOFTWARE AND PRODUCTS

The Well: scientific-simulation evaluation

2026.07 · Scientific machine learning and evaluation

Working research and merged software contributions

AI for Science · AI-assisted

Studies denominator sensitivity in normalized rollout metrics, physical-unit error comparisons and the inverse problem of identifying active-matter dynamics from aggregate observations.

Software contributions address epsilon propagation in VRMSE/VMSE (#92, merged 23 July 2026) and long-horizon batch averaging (#93, merged 12 October 2026). AI-assisted development is disclosed. A merged fix is distinct from a released package or a benchmark-performance claim.

Methods, related work and links ↗

Language-model calibration and statistical identification

2026 · AI evaluation and mathematical statistics

Unpublished empirical and theoretical research

AI for Science · AI-assisted

An empirical line examines whether hidden representations add information about correctness beyond output-based baselines. A Qwen2.5-0.5B / ARC-Challenge pilot evaluated 1,165 held-out test questions. The AUROC increment was 0.00218, with a paired-bootstrap 95% interval of −0.01857 to 0.02283: no clear improvement in that configuration.

A distinct mathematical line studies finite exchangeability, deletion models, expected-correctness summaries and statistical identification. It distinguishes what full observations identify from what rank or aggregate summaries retain, including examples of positive population ambiguity. These draft results are not journal acceptances or a general negative claim about model internals.

Methods, related work and links ↗

A Density-Matrix Belief-State World Model

2026 — · Prediction and scientific software Research prototypes

AI for Science · AI-assisted

A programme within Omytea studying density-matrix belief states as representations of uncertainty over multiple futures. The work uses classically executed open-system mathematics, operator/channel updates and records of how observations revise a forecast.

Current prototypes study calibration and auditable updates. They do not establish forecasting superiority or quantum-computing advantage. Related work includes the Personal Future Console and a Python package for density-matrix and Lindblad-style calculations.

Methods, related work and links ↗

Omytea

2026 — · Founder · product and research in development

Omytea is building a human–agent internet: personal intelligence shaped through permitted life records and interaction, with portable memory, private agent communication and optional shared intermediaries. The network design applies blockchain principles to signed, linked records and independently verifiable cooperation. Local cultivation and signing prototypes exist; portable memory, continuous mobile recording and the wider network remain in development. Predictive world models, embodied interfaces and AI-for-science research support the longer programme.

omyteaai.com ↗ · Research and public instruments

Related instruments include the Personal Future Console and classically executed density-matrix scientific software. GolfCal-1k remains a calibration/evaluation specification with synthetic prototype data; its name is a target, not evidence of 1,000 real shots.

Swindo

2026 — · Product in development

Video-based golf-motion analysis combining monocular pose reconstruction, body profiles, calibration and motion review. The pipeline infers clubhead position from the body/arm model rather than directly detecting it. Related research examines inertial sensing and the limits of Wi-Fi/RF motion inference.

Project and limitations ↗

Nye Clock · 弍时仪

A celestial sexagenary instrument combining a date-driven Sun–Earth–Moon orrery with the heavenly-stem and earthly-branch cycle. The portfolio also contains a scientific sky, star atlas and geographic footprint instruments.

Open instrument ↗

Inference Rent in Generative AI: Intellectual Monopoly and Metered Access to Non-Rival Capability

2026.01–2026.07 · Political economy of technology

Public SSRN preprint · revised September 2026

Develops inference rent as a category for analysing metered access to generative-AI capability: an underlying non-rival capability is made available through priced inference and platform-controlled access.
Distinguishes technological capability, access conditions and the institutional power to collect rents. The September 2026 revision develops the earlier technological-monopoly-rent manuscript rather than constituting a separate new paper.

Methods, related work and links ↗

Frontier AI and the Limits of Monopoly Self-Correction: Inference Rent, Compute Bottlenecks, and Platform Lock-In

2026.01–2026.07 · Political economy and industrial organisation Public SSRN preprint

Examines why frontier-AI markets may fail to correct concentrated power through ordinary competitive entry, connecting inference rent, compute bottlenecks and platform lock-in.
A conceptual political-economy paper, distinct from the documentary study of public AI-compute programmes. Its public-preprint status does not imply journal acceptance.

Methods, related work and links ↗

Camera, Player, Engine: An Epistemology for Writing About a System You Are Inside

2026.01–2026.07 · Epistemology and social inquiry Public SSRN preprint

Develops a framework for writing about systems in which the writer is also a participant: observation, strategic participation and intervention create different epistemic demands.
The camera, player and engine distinguish descriptive distance, situated action and system-changing activity without treating any one stance as a neutral view from outside.

Methods, related work and links ↗

An Invitation After Abundance: Reading the Present, Wagering the Future

2026.01–2026.07 · Political economy · 1,162 pages Self-published book · 2026

A political-economy volume: reading the present and wagering the future — technology, power, and institutions after abundance.
A parallel original to the Chinese-language 和平请柬 — two independent works.
ISBN 979-8-9960105-0-9 (ebook) · 979-8-9960105-2-3 (paperback) · 979-8-9960105-3-0 (hardcover).
A parallel original work: the English and Chinese books are independently written, not translations of one another.

Methods, related work and links ↗

和平请柬

2026.01–2026.07 · Chinese-language political-economy monograph · 1,068 pages

Self-published book · 2026

A Chinese-language monograph written as an invitation: at the threshold of leaving university, an attempt to read the age being entered and judge where it is going — productive capacity outrunning our institutional language, and the question of who pays for that gap.
A parallel original to An Invitation After Abundance — two independent works. Hardcover · ISBN 979-8-9960105-1-6.
A parallel original work: the English and Chinese books are independently written, not translations of one another.

Methods, related work and links ↗

Sovereign Scintillation Over Grand Fluctuation: An Observer-State Theory of Physics

2024.05–2024.08 · Foundations of physics · book draft Unpublished book manuscript

A single-volume framework: observer self-states plus a universal-prior-weighted transition rule as primitives, asking what physics — locality, modular flow, the Born rule — does and does not follow.

Methods, related work and links ↗

Additional essay drafts include The Abundance That Isn’t and The Rent Machine. These are writing projects, without a claim of magazine acceptance.

CMS Jet Physics

2025.05–2025.09 · Roloff group · Brown University Graduate research

Graduate research on CMS jet physics using PYTHIA simulations to study parton-shower cascades and jet structure.
Examined how jet topology and simulation choices inform the interpretation of beyond-the-Standard-Model signatures. This experience concerns simulation and analysis, without a claim of a current CMS appointment or detector-operation role.

Methods, related work and links ↗

Unified Theory for 3ω Thermal Characterization: Finite Heater Length and Environmental Coupling

2025.10–2026.02 · Kuehne group · Brown University Graduate research

Participated in thermal-transport research on carbon-nanotube 3ω characterization and finite-heater impedance modelling.
The work considers heater length, environmental coupling, heater heat capacity and boundary resistance across vacuum, fluid and supported configurations. Participation in the research and manuscript work is distinct from confirmed public-paper authorship.

Methods, related work and links ↗

MAXI J1348-630 Black-Hole X-ray Binary

2023.09–2024.05 · X-ray astrophysics · Shandong University Completed undergraduate thesis

Undergraduate thesis on MAXI J1348-630 using public NICER and NuSTAR observations. The analysis used HEASoft/NuSTARDAS, source/background extraction and NuSTAR FPMA/FPMB products.
Timing work used Powspec, while spectral analysis used XSPEC and model-residual comparisons. The examined NuSTAR data did not yield a useful QPO identification; fitting residuals and model limitations remained part of the interpretation.

Methods, related work and links ↗

TOI Stellar-Flare Spectroscopy

2022.04–2023.04 · Stellar spectroscopy · Shandong University

Completed undergraduate research

Studied TOI stellar-flare spectroscopy and the interpretation of time-dependent stellar activity.
Connected flare spectral behaviour to planetary radiation environments and habitability questions, with the observational and interpretive scope stated separately.

Methods, related work and links ↗

In-Beam γ-ray Spectroscopy

2022 · Nuclear spectroscopy · Shandong University Completed undergraduate research

Analysed in-beam γ-ray spectra from α-induced nuclear reactions, using energy-matrix gates to distinguish coincidence structures and identify residual nuclides.
The experience focused on spectral interpretation and nuclide identification; a separate accelerator visit is not presented as conducting an irradiation experiment.

Methods, related work and links ↗

Axion-Like Dark Matter

2022.03–2022.05 · Dark-matter phenomenology · Beijing Normal University

Completed research project

Studied axion-like dark-matter phenomenology through photon survival probabilities and their dependence on candidate-particle parameters.
The project developed parameter-space interpretation rather than a new experimental exclusion limit.

Methods, related work and links ↗

08 METHODS AND WORKING TOOLS

Scientific-computing and AI-assisted research workflows; simulation diagnostics, statistical calibration, exact finite-model checks, reproducible analysis and manuscript preparation. Project pages identify the particular tools and validation scope. Earlier observational work used HEASoft, NuSTARDAS, Powspec and XSPEC; jet studies used PYTHIA.

Research and writing in English and Chinese. Programming/tool-assisted outputs are distinguished from independent conceptual understanding and real-world validation.

10 CONTACT

Open to research & collaboration.

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